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Acrylonitrile Electrohydrodimerization (EHD) — 1D Planar Electrode Model

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Scope: 1D Nernst diffusion layer (δ), Nernst–Planck transport with migration, Tafel kinetics for ADPN/PN/HER, phosphate buffer chemistry, local-equilibrium multiphase with arithmetic-mean $D_{i,mix}$, on a Cd cathode. **Operating temperature: $T = 298.15\text{ K}$ ($25\text{ }^{\circ}\text{C}$).**

Convention

All concentrations c_i in this guide are **aqueous-phase** values [mol m^{-3} of aqueous solution]. Under the local-equilibrium assumption ($Da \gg 1$), organic and aqueous phases equilibrate instantaneously at every position: $c_{i,org}(x) = m_i \times c_{i,aq}(x)$. The organic

phase acts as a parallel transport pathway captured entirely through the effective diffusivity $D_{i,mix}$. **No explicit phase transfer term R_{PT} appears** in the governing equations. No volume-fraction prefactors on any source terms.

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1. Physical Domain

The model solves steady-state species transport across a stagnant Nernst diffusion layer of thickness δ [m] adjacent to a planar cadmium cathode (at $x = 0$), with the well-mixed bulk electrolyte at $x = \delta$. Framework: Corpus et al. (Joule 2023) and Weng, Bell & Weber (PCCP 2018).

The diffusion layer contains dispersed organic droplets characterised by their volume fraction ϵ_{org} [—]. Under the local-equilibrium assumption, the organic and aqueous phases are in instantaneous

equilibrium everywhere — the droplets provide a parallel diffusion pathway for species that partition into the organic phase. All electrochemical reactions occur at the electrode surface and enter the model as flux boundary conditions at $x = 0$. H_2 desorbs immediately and is not tracked.

ELECTRODE (Cd)	DIFFUSION LAYER (δ)	BULK ELECTROLYTE
$x = 0$	organic droplets in local equil.	$x = \delta$
Tafel kinetics	NP transport with migration	Dirichlet BCs
ADPN + PN + HER	$D_{i,mix}(\epsilon_{org})$ eff. diffusivity	$c_i = c_{i,bulk}$
Flux BCs	Phosphate buffer chemistry	
	No R_{PT} (local equilibrium)	
	Current conservation (ϕ_l)	

Key findings from Bloomquist et al. (CEJ 2026, 528, 172125):

- FE_ADPN increases from <50% to >80% when ϵ_{org} exceeds the solubility limit (~ 0.09)
- Bubble-induced convection dominates over inlet flow regime
- High selectivity maintained at $j > 200 \text{ mA cm}^{-2}$
- ϵ_{org} explored from 0.05 to 0.30

2. Species and Degrees of Freedom

The model tracks 9 dissolved species. Eight are independent unknowns; Na^+ is computed from electroneutrality.

Index	Species	Charge z_i	Role
1	H^+	+1	Proton
2	OH^-	-1	Hydroxide (product of all 3 cathodic reactions)
3	$H_2PO_4^-$	-1	Buffer
4	HPO_4^{2-}	-2	Buffer
5	PO_4^{3-}	-3	Buffer
6	AN ($CH_2=CHCN$)	0	Acrylonitrile (reactant)
7	ADPN ($NC(CH_2)_4CN$)	0	Adiponitrile (product)
8	PN (CH_3CH_2CN)	0	Propionitrile (byproduct)
9	Na^+	+1	From electroneutrality (not a DOF)

TBA⁺ treatment: Tetrabutylammonium ($0.02 \text{ M} = 20 \text{ mol m}^{-3}$) is present as a selectivity agent but is lumped into the Na^+ background — both carry $z = +1$, and since Na^+ is recovered from electroneutrality rather than transported, no mobility mismatch

enters the model. The effective background Na^+ input is $c_{\text{Na,input}} = 3 \times 500 + 20 = 1520 \text{ mol m}^{-3}$ (see §6.4).

Na^+ is determined algebraically from local electroneutrality:

$$c_{\text{Na}} = c_{\text{OH}} + c_{\text{H}_2\text{PO}_4} + 2 \cdot c_{\text{HPO}_4} + 3 \cdot c_{\text{PO}_4} - c_{\text{H}}$$

Each finite-volume cell carries **9 degrees of freedom**: 8 log-concentrations (species 1–8) plus the electrolyte potential ϕ_l [V]. Log-concentrations are defined as $u_i = \ln(c_i)$ and recovered via $c_i = \exp(\text{clamp}(u_i, -50, 50))$. With $N_{\text{mesh}} = 100$: **900 total unknowns**.

3. Governing Equations

3.1 Nernst–Planck Transport with Migration

For each species i at steady state:

$$0 = \frac{\partial}{\partial x} \left[D_{i,\text{mix}} \frac{\partial c_i}{\partial x} + \frac{z_i F}{RT} D_{i,\text{mix}} c_i \frac{\partial \phi_l}{\partial x} \right] + R_{\text{buf},i}$$

$$0 = d/dx [D_{i,\text{mix}} \times dc_i/dx + (z_i \cdot F)/(R \cdot T) \times D_{i,\text{mix}} \times c_i \times d\phi_l/dx] + R_{\text{buf},i}$$

Symbol	Definition	Unit
c_i	Aqueous-phase concentration of species i	mol m^{-3}
$D_{i,\text{mix}}$	Effective diffusivity through emulsion (§4)	$\text{m}^2 \text{s}^{-1}$
z_i	Charge number	—
F	Faraday constant = 96,485.332	C mol^{-1}
R	Gas constant = 8.314463	$\text{J mol}^{-1} \text{K}^{-1}$
T	Temperature = 298.15 K (25 °C)	K
ϕ_l	Electrolyte potential (solved DOF)	V
$R_{\text{buf},i}$	Buffer reaction source for species i (§6)	$\text{mol m}^{-3} \text{s}^{-1}$
x	Spatial coordinate (0 = electrode, δ = bulk)	m
δ	Diffusion layer thickness	m

No R_{PT} term. Under the local-equilibrium assumption, the organic and aqueous phases equilibrate instantaneously and there is no driving force for interphase transfer. All multiphase effects are captured through $D_{i,\text{mix}}$.

No prefactor on R_{buf} . Buffer reactions occur in the aqueous phase, the rate law uses aqueous concentrations, and c_i is an aqueous concentration. No correction is needed.

3.2 Current Conservation (Potential Equation)

The 9th equation per cell enforces zero net ionic current in the diffusion layer. No external current flows through the solution between $x = 0$ and $x = \delta$ — current enters and exits only at the electrode. This constraint determines the electrolyte potential ϕ_l at every position:

$$\frac{\partial}{\partial x} \left[\sum_i z_i N_i \right] = 0$$

where N_i is the Nernst–Planck flux of species i and the sum runs over charged species only. In the FV discretisation, this becomes: the sum of ionic SG fluxes at the right face of each cell must equal the sum at the left face.

Well-posedness (gauge fix): The current-conservation equation $d/dx [\sum z_i N_i] = 0$ is invariant under a uniform shift of ϕ_l — the system has one null direction. The Dirichlet condition $\phi_l(\delta) = 0$ (§7.2) kills this null direction and makes the Jacobian non-singular. Without it, Newton cannot converge.

3.3 Scharfetter–Gummel Flux Discretisation

Standard centred-difference flux approximations become numerically unstable when the dimensionless migration Péclet number $\alpha = z_i F \Delta\phi / (RT)$ exceeds ~ 2 per cell, producing spurious oscillations. The SG scheme uses an exponential fitting to handle arbitrarily large α stably. For neutral species ($\alpha = 0$) it reduces to the standard centred difference:

$$J_i = -\frac{D_{i,\text{mix}}}{\Delta x} [B(\alpha) c_R - B(-\alpha) c_L], \quad \alpha = \frac{z_i F (\phi_R - \phi_L)}{RT}$$

with the Bernoulli functions:

$$B(\alpha) = \frac{\alpha}{e^\alpha - 1}, \quad B(-\alpha) = \frac{\alpha e^\alpha}{e^\alpha - 1}$$

where c_L, c_R are aqueous concentrations in the left and right cells [mol m^{-3}]; ϕ_L, ϕ_R are electrolyte potentials [V]; and Δx is the distance between cell centres [m]. α is clamped to $[-700, 700]$ to avoid floating-point overflow.

function sg_flux($c_L, c_R, \phi_L, \phi_R, D_{\text{mix}}, z_i, dx, T=298.15$)

```

    z_i == 0 && return -D_mix * (c_R - c_L) / dx
    alpha = clamp(z_i * F * (phi_R - phi_L) / (R * T), -700.0, 700.0)
    abs(alpha) < 1e-10 && return -D_mix * (c_R - c_L) / dx
    B_pos = alpha / (exp(alpha) - 1.0)
    B_neg = alpha * exp(alpha) / (exp(alpha) - 1.0)

```

```

    return -D_mix / dx * (B_pos * c_R - B_neg * c_L)
end

```

4. Mixture-Averaged Diffusivities

4.1 Arithmetic Mean (Current Formulation)

The effective diffusivity is a simple volume-weighted arithmetic mean of the diffusivity in each phase:

$$D_{i,\text{mix}} = \varepsilon_{\text{org}} D_{i,\text{org}} + (1 - \varepsilon_{\text{org}}) D_{i,\text{aq}}$$

Symbol	Definition	Unit
ε_{org}	Organic-phase volume fraction (sweep parameter)	—
$D_{i,\text{aq}}$	Diffusivity of species i in pure aqueous phase	$\text{m}^2 \text{s}^{-1}$
$D_{i,\text{org}}$	Diffusivity of species i in the organic phase	$\text{m}^2 \text{s}^{-1}$

This is the simplest mixing rule and does **not** include the partition coefficient m_i . The full local-equilibrium effective diffusivity is $D_{i,\text{eff}} = (1 - \varepsilon_{\text{org}}) D_{i,\text{aq}} + \varepsilon_{\text{org}} m_i D_{i,\text{org}}$ (see §4.2); the arithmetic mean is the first thing to test and will be upgraded if it underpredicts the experimental FE_{ADPN} enhancement.

Ions are insoluble in the organic phase, so $D_{i,\text{org}} = 0$ for all ions:

$$D_{i,\text{mix}}^{\text{ion}} = (1 - \varepsilon_{\text{org}}) D_{i,\text{aq}}$$

This means increasing ε_{org} **reduces ionic transport**, creating an ohmic penalty. Neutral organic diffusivities are estimated via the Stokes–Einstein viscosity ratio: $D_{i,\text{org}} \approx D_{i,\text{aq}} \times (\mu_{\text{aq}}/\mu_{\text{org}}) \approx 2.6 \times D_{i,\text{aq}}$ ($\mu_{\text{H}_2\text{O}} = 0.89 \text{ mPa}\cdot\text{s}$, $\mu_{\text{AN}} = 0.34 \text{ mPa}\cdot\text{s}$ at 25°C).

Species	$D_{\text{aq}} (\text{m}^2/\text{s})$	$D_{\text{org}} (\text{m}^2/\text{s})$	Notes
AN	2.30×10^{-9}	6.00×10^{-9}	Stokes–Einstein $\times 2.6$
ADPN	1.50×10^{-9}	3.90×10^{-9}	Stokes–Einstein $\times 2.6$
PN	2.30×10^{-9}	6.00×10^{-9}	Stokes–Einstein $\times 2.6$
H^+	9.31×10^{-9}	0	Insoluble in organic
OH^-	5.27×10^{-9}	0	Insoluble in organic
H_2PO_4^-	0.846×10^{-9}	0	Insoluble in organic

Species	D _{aq} (m ² /s)	D _{org} (m ² /s)	Notes
HPO ₄ ²⁻	0.690 × 10 ⁻⁹	0	Insoluble in organic
PO ₄ ³⁻	0.610 × 10 ⁻⁹	0	Insoluble in organic
Na ⁺	1.33 × 10 ⁻⁹	0	Insoluble in organic

D_{aq} from Suwanvaipattana et al. (2017) and CRC Handbook. All values at T = 25 °C.

```
const T = 298.15 # K, operating temperature
```

```
const D_aq = [9.31e-9, 5.27e-9, 0.846e-9, 0.69e-9, 0.61e-9, # H+ OH- H2PO4- HPO4 2- PO4 3-
              2.3e-9, 1.5e-9, 2.3e-9] # AN ADPN PN
```

```
const D_org = [0.0, 0.0, 0.0, 0.0, 0.0, # ions: insoluble
              6.0e-9, 3.9e-9, 6.0e-9] # AN ADPN PN
```

```
D_mix(i, eps_org) = eps_org * D_org[i] + (1.0 - eps_org) * D_aq[i]
```

Effect of ϵ_{org} on Mixture Diffusivities

ϵ_{org}	D _{AN,mix} (m ² /s)	D _{OH,mix} (m ² /s)	D _{AN} /D _{AN,eq}	D _{OH} /D _{OH,eq}
0.00	2.30 × 10 ⁻⁹	5.27 × 10 ⁻⁹	1.00	1.00
0.09	2.63 × 10 ⁻⁹	4.80 × 10 ⁻⁹	1.14	0.91
0.15	2.86 × 10 ⁻⁹	4.48 × 10 ⁻⁹	1.24	0.85
0.25	3.23 × 10 ⁻⁹	3.95 × 10 ⁻⁹	1.40	0.75
0.30	3.41 × 10 ⁻⁹	3.69 × 10 ⁻⁹	1.48	0.70

AN transport improves by 48% at $\epsilon_{\text{org}} = 0.30$ while OH⁻ transport degrades by 30% — this quantifies the selectivity–conductivity trade-off.

4.2 m_i-Corrected Upgrade (Future)

If the arithmetic mean is too weak to reproduce the experimental FE enhancement, the physically correct local-equilibrium effective diffusivity includes the partition coefficient $m_i = c_{i,\text{org}}/c_{i,\text{aq}}$:

$$D_{i,\text{eff}} = (1 - \epsilon_{\text{org}}) D_{i,\text{aq}} + \epsilon_{\text{org}} m_i D_{i,\text{org}}$$

For AN ($m_{\text{AN}} = 11.59$) at $\epsilon_{\text{org}} = 0.25$: $D_{\text{AN,eff}} = 19.1 \times 10^{-9}$ m²/s vs $D_{\text{AN,mix}} = 3.23 \times 10^{-9}$. For ions: $m_i = 0$, so $D_{\text{ion,eff}} = (1 - \epsilon_{\text{org}}) D_{i,\text{aq}}$ — no change.

One-line switch from arithmetic to m_i-corrected:

```
D_mix(i, eo) = eo * D_org[i] + (1-eo) * D_aq[i]      # arithmetic
D_eff(i, eo) = eo * m[i] * D_org[i] + (1-eo) * D_aq[i]  # m-corrected
```

5. Electrochemical Kinetics

5.1 Competing Cathodic Reactions

Reaction	Stoichiometry	n_e
R1: Electrohydrodimerization	$2 \text{ AN} + 2 \text{ H}_2\text{O} + 2\text{e}^- \rightarrow \text{ADPN} + 2 \text{ OH}^-$	2
R2: Hydrogenation	$\text{AN} + 2 \text{ H}_2\text{O} + 2\text{e}^- \rightarrow \text{PN} + 2 \text{ OH}^-$	2
R3: Hydrogen evolution	$2 \text{ H}_2\text{O} + 2\text{e}^- \rightarrow \text{H}_2 + 2 \text{ OH}^-$	2

All three reactions transfer $n_e = 2$ electrons and produce 2 OH^- .

5.2 Tafel Rate Expressions

All reactions operate far from equilibrium; the cathodic Tafel approximation applies. The current densities j_r [A m^{-2}] at the electrode surface are:

R1 — ADPN (second-order in AN):

$$j_1 = j_{0,1} \left(\frac{c_{\text{AN}}}{c_{\text{ref}}} \right)^2 \exp \left(-\frac{\alpha_{c,1} F \eta_1}{RT} \right)$$

$$j_1 = j_{0,1} \times (c_{\text{AN}} / c_{\text{ref}})^2 \times \exp(-\alpha_{c,1} \times F \times \eta_1 / (R \times T))$$

R2 — PN (first-order in AN):

$$j_2 = j_{0,2} \frac{c_{\text{AN}}}{c_{\text{ref}}} \exp \left(-\frac{\alpha_{c,2} F \eta_2}{RT} \right)$$

$$j_2 = j_{0,2} \times (c_{\text{AN}} / c_{\text{ref}}) \times \exp(-\alpha_{c,2} \times F \times \eta_2 / (R \times T))$$

R3 — HER (independent of AN):

$$j_3 = j_{0,3} \exp \left(-\frac{\alpha_{c,3} F \eta_3}{RT} \right)$$

$$j_3 = j_{0,3} \times \exp(-\alpha_{c,3} \times F \times \eta_3 / (R \times T))$$

The overpotential for each reaction:

$$\eta_r = (\phi_s - \phi_l) - E_r^0$$

Symbol	Definition	Unit
j_r	Current density for reaction r (positive = cathodic)	A m^{-2}
$j_{0,r}$	Exchange current density for reaction r	A m^{-2}
c_{AN}	Aqueous AN concentration at the electrode surface	mol m^{-3}
c_{ref}	Reference concentration = $1,000 \text{ mol m}^{-3}$ (= 1 M)	mol m^{-3}
$\alpha_{c,r}$	Cathodic transfer coefficient for reaction r	—
η_r	Overpotential for reaction r	V
ϕ_s	Applied cathode potential	V vs SHE
ϕ_l	Solved electrolyte potential at the electrode	V
E_r^0	Standard reduction potential for reaction r	V vs SHE

Unit note: $j_{0,r}$ is given in the fitting table (§9.2) in mA cm^{-2} . Convert to SI via $\times 10$: $1 \text{ mA cm}^{-2} = 10 \text{ A m}^{-2}$. $c_{\text{ref}} = 1000 \text{ mol m}^{-3}$ so that $(c_{\text{AN}}/c_{\text{ref}})$ is dimensionless; c_{AN} can exceed c_{ref} at high AN loading (above solubility, $c_{\text{AN}} \approx 1310 \text{ mol m}^{-3}$), which is physical.

The c^2 vs c dependence is the central selectivity feature: as c_{AN} drops at the electrode under high current density, ADPN selectivity falls faster than PN.

The Faradaic efficiency for ADPN:

$$\text{FE}_{\text{ADPN}} = \frac{j_1}{j_1 + j_2 + j_3} \times 100\%$$

5.3 Alternative: Huang/Suwanvaipattana Rate Law

Huang et al. (CEJ 2020) use a Langmuir-adsorption formulation on Pb:

$$r_1 = k_1 \left[\frac{K_{\text{AN}} c_{\text{AN}}}{1 + K_{\text{AN}} c_{\text{AN}}} \right]^2, \quad r_2 = k_2 \frac{K_{\text{AN}} c_{\text{AN}}}{1 + K_{\text{AN}} c_{\text{AN}}}$$

Fitted parameters (on Pb): $k_{1,\text{ref}} = 0.56 \pm 0.23 \text{ h}^{-1} \text{ m}$, $k_{2,\text{ref}} = 0.075 \pm 0.029 \text{ L mol}^{-1} \text{ h}^{-1} \text{ m}$, $K_{\text{AN},\text{ref}} = 10.80 \pm 2.38 \text{ L mol}^{-1}$, $\Delta H_{\text{AN}} = -7.15 \text{ kJ mol}^{-1}$, $\alpha_1 = 1.48 \text{ V}^{-1}$, $\alpha_2 = 2.04 \text{ V}^{-1}$.

6. Phosphate Buffer Chemistry

6.1 Equilibria

Three equilibria are tracked. H_3PO_4 dissociation ($\text{pK} = 2.12$) is omitted because it is negligible at the operating pH of 12–13.

#	Reaction	K (25°C, mol/m ³ units)	pK
1	$\text{H}_2\text{O} \rightleftharpoons \text{H}^+ + \text{OH}^-$	$K_w = 1.0 \times 10^{-8} \text{ (mol/m}^3\text{)}^2$	14.00
2	$\text{H}_2\text{PO}_4^- \rightleftharpoons \text{H}^+ + \text{HPO}_4^{2-}$	$K_{a2} = 6.2 \times 10^{-5} \text{ mol/m}^3$	7.21
3	$\text{HPO}_4^{2-} \rightleftharpoons \text{H}^+ + \text{PO}_4^{3-}$	$K_{a3} = 4.8 \times 10^{-10} \text{ mol/m}^3$	12.32

Unit convention: All K values are in mol m⁻³ (SI). Relation to standard mol L⁻¹ values: $K_w = 10^{-14} \text{ (mol/L)}^2 \times (1000)^2 = 10^{-8} \text{ (mol/m}^3\text{)}^2$. The pK values listed are in standard units ($-\log_{10}[\text{mol/L}]$).

Reverse rate constants are always derived: $k_r = k_f / K_{eq}$. Never set independently.

6.2 R_buf,i Expressions

For each equilibrium, the net rate [mol m⁻³ s⁻¹] is forward minus reverse:

$$\begin{aligned}
 r_1 &= k_{1,f} - k_{1,r} c_{\text{H}^+} c_{\text{OH}^-} \\
 r_2 &= k_{2,f} c_{\text{H}_2\text{PO}_4^-} - k_{2,r} c_{\text{H}^+} c_{\text{HPO}_4^{2-}} \\
 r_3 &= k_{3,f} c_{\text{HPO}_4^{2-}} - k_{3,r} c_{\text{H}^+} c_{\text{PO}_4^{3-}}
 \end{aligned}$$

Species allocation:

$$\begin{aligned}
 R_{\text{buf}}[\text{H}^+] &= +r_1 + r_2 + r_3 & R_{\text{buf}}[\text{OH}^-] &= +r_1 \\
 R_{\text{buf}}[\text{H}_2\text{PO}_4^-] &= -r_2 & R_{\text{buf}}[\text{HPO}_4^{2-}] &= +r_2 - r_3 & R_{\text{buf}}[\text{PO}_4^{3-}] &= +r_3
 \end{aligned}$$

$R_{\text{buf}} = 0$ for AN, ADPN, and PN (no buffer reactions). **No prefactor** — c_i is aqueous; R_{buf} is rate per aqueous volume.

function buffer_sources!(R, c_H, c_OH, c_H2P04, c_HP04, c_P04, alpha_buf=1.0)

```

r1 = k1f - k1r * c_H * c_OH           #  $\text{H}_2\text{O} \rightleftharpoons \text{H}^+ + \text{OH}^-$ 
r2 = k2f * c_H2P04 - k2r * c_H * c_HP04 #  $\text{H}_2\text{PO}_4^- \rightleftharpoons \text{H}^+ + \text{HPO}_4^{2-}$ 
r3 = k3f * c_HP04 - k3r * c_H * c_P04   #  $\text{HPO}_4^{2-} \rightleftharpoons \text{H}^+ + \text{PO}_4^{3-}$ 
R[1] = alpha_buf * (+r1 + r2 + r3)      #  $\text{H}^+$ 
R[2] = alpha_buf * (+r1)                 #  $\text{OH}^-$ 
R[3] = alpha_buf * (-r2)                 #  $\text{H}_2\text{PO}_4^-$ 

```

```

R[4] = alpha_buf * (+r2 - r3)      #  $HPO_4^{2-}$ 
R[5] = alpha_buf * (+r3)          #  $PO_4^{3-}$ 
R[6:8] .= 0.0                     # AN, ADPN, PN
end

```

The optional `alpha_buf` argument (default 1.0) is the buffer continuation parameter used during bootstrap (see §10.6, §12).

6.3 Forward Rate Constants

k	Value	Unit
$k_{1,f}$	2.4×10^{-5}	s^{-1}
$k_{2,f}$	$\sim 10^6$	s^{-1}
$k_{3,f}$	$\sim 10^2$	s^{-1}

6.4 Bulk Equilibrium Initial Guess

This is the most critical step for convergence. Buffer reactions are stiff: forward and reverse rates are both large but nearly cancel at equilibrium. An initial guess that is even slightly off-equilibrium produces buffer source terms of $O(10^3\text{--}10^5) \text{ mol m}^{-3} \text{ s}^{-1}$ — orders of magnitude larger than transport fluxes — causing Newton to fail or stall for hundreds of iterations even at $\alpha_{\text{buf}} = 0.001$.

The fix: solve the phosphate charge balance exactly before starting Newton.

The equilibrium speciation follows from two constraints:

Phosphate conservation:

$$c_{H_2PO_4} + c_{HPO_4} + c_{PO_4} = C_{P,\text{total}}$$

Expressing each species in terms of c_H alone:

$$c_{HPO_4} = \frac{C_{P,\text{total}}}{c_H/K_{a2} + 1 + K_{a3}/c_H}, \quad c_{H_2PO_4} = \frac{c_H c_{HPO_4}}{K_{a2}}, \quad c_{PO_4} = \frac{K_{a3} c_{HPO_4}}{c_H}$$

Charge balance (one equation in c_H):

$$c_{Na,\text{input}} + c_H = \frac{K_w}{c_H} + c_{H_2PO_4} + 2 c_{HPO_4} + 3 c_{PO_4}$$

where $c_{Na,\text{input}} = 3 \times C_{Na_3PO_4} + C_{TBA-OH} = 3 \times 500 + 20 = \mathbf{1520 \text{ mol m}^{-3}}$ (TBA⁺ lumped into Na⁺, see §2).

This is a 1D root-finding problem in $\log_{10}(c_H)$:

```

const K_w  = 1.0e-8    # (mol/m3)2, water autoprotolysis at 25°C
const K_a2 = 6.2e-5    # mol/m3, H2PO4- ↔ H+ + HPO42- at 25°C
const K_a3 = 4.8e-10   # mol/m3, HPO42- ↔ H+ + PO43- at 25°C

"""
    solve_phosphate_equilibrium(C_P_total, c_Na_input) → NamedTuple

```

Solve the phosphate buffer charge balance in the well-mixed bulk.
All concentrations in mol m⁻³.

```

C_P_total : total phosphate [mol/m3] (0.5 M Na3PO4 → 500)
c_Na_input : total monovalent cation input [mol/m3]
              (3×500 Na+ from Na3PO4 + 20 TBA+ from TBA-OH → 1520)

```

Returns NamedTuple (H, OH, H2P04, HP04, P04, Na, pH).

```

"""
function solve_phosphate_equilibrium(C_P_total=500.0, c_Na_input=1520.0)

```

```

    function charge_balance(log10_cH)
        cH    = 10.0^log10_cH
        cOH    = K_w / cH
        denom = cH / K_a2 + 1.0 + K_a3 / cH
        cHP04  = C_P_total / denom
        cH2P04 = cH * cHP04 / K_a2
        cP04   = K_a3 * cHP04 / cH
        # charge balance: (fixed cations + H+) - anions = 0
        return c_Na_input + cH - (cOH + cH2P04 + 2.0*cHP04 + 3.0*cP04)
    end

```

```

    # Bisect over pH 6–15 in standard units

```

```

    # → log10(c_H [mol/m3]) ∈ [-12, -3]

```

```

    using Roots

```

```

    log10_cH_eq = find_zero(charge_balance, (-12.0, -3.0), Bisection())

```

```

    cH    = 10.0^log10_cH_eq
    cOH    = K_w / cH
    denom = cH / K_a2 + 1.0 + K_a3 / cH
    cHP04  = C_P_total / denom
    cH2P04 = cH * cHP04 / K_a2

```

```

cP04    = K_a3 * cHP04 / cH
pH_std = -log10(cH / 1000.0)  # pH in standard mol L-1 units

return (H=cH, OH=cOH, H2P04=cH2P04, HP04=cHP04, P04=cP04,
        Na=c_Na_input, pH=pH_std)
end

```

Verification — always run before starting Newton:

```
c_eq = solve_phosphate_equilibrium()
```

```
# Equilibrium product checks
```

```

@assert abs(c_eq.H * c_eq.OH - K_w) / K_w < 1e-8 "K_w mismatch"
@assert abs(c_eq.H * c_eq.HP04 / c_eq.H2P04 - K_a2) / K_a2 < 1e-8 "K_a2 mismatch"
@assert abs(c_eq.H * c_eq.P04 / c_eq.HP04 - K_a3) / K_a3 < 1e-8 "K_a3 mismatch"

```

```
# Buffer sources must be zero at equilibrium
```

```
R_check = zeros(8)
```

```
buffer_sources!(R_check, c_eq.H, c_eq.OH, c_eq.H2P04, c_eq.HP04, c_eq.P04)
```

```
@assert maximum(abs.(R_check)) < 1e-6 "Buffer not at equilibrium: max|R_buf| = $(maximum(abs.(R_check)))"
```

Expected equilibrium values for 0.5 M Na₃PO₄ + 0.02 M TBA-OH at 25°C:

Species	Approximate value	Unit
c _H	~10 ⁻¹⁰	mol m ⁻³
c _{OH}	~100	mol m ⁻³
c _{H₂PO₄}	~0	mol m ⁻³
c _{HPO₄}	~70–90	mol m ⁻³
c _{PO₄}	~410–430	mol m ⁻³
c _{Na}	1520	mol m ⁻³
pH	~13.0	—

Most phosphate sits as PO₄³⁻ since pH ≈ 13.0 is well above pKa3 = 12.32.

Building the full DOF initial guess vector:

```
# DOF layout: 8 log-concentrations + 1 ϕl per cell, cell-major (interleaved)
```

```
# Species order: H+(1), OH-(2), H2PO4-(3), HPO42-(4), PO43-(5), AN(6), ADPN(7), PN(8)
```

```
function make_initial_guess(N_mesh, c_eq, c_AN_bulk; c_seed=1e-3)
```

```
    u0 = zeros(9 * N_mesh)
```

```
    log_c0 = [log(c_eq.H),      log(c_eq.OH),
```

```

        log(c_eq.H2P04), log(c_eq.HP04), log(c_eq.P04),
        log(c_AN_bulk),
        log(c_seed),      # ADPN: small seed to avoid log(0)
        log(c_seed)]      # PN: small seed to avoid log(0)
    for ix in 1:N_mesh
        u0[9*(ix-1)+1 : 9*(ix-1)+8] .= log_c0
        u0[9*(ix-1)+9] = 0.0 #  $\phi_l = 0$  everywhere
    end
    return u0
end

```

With this initialisation and $\alpha_{\text{buf}} = 0$ (buffer off), $\alpha_{\text{kin}} = 0$ (kinetics off), the residual is **identically zero**: no concentration gradients, no current, no buffer sources. Newton converges in 0 iterations — the ideal starting point for the bootstrap protocol (§12).

7. Boundary Conditions

7.1 Electrode ($x = 0$): Faradaic Flux

The species fluxes at the electrode surface are set by Faraday's law applied to the three cathodic reactions. Two AN molecules are consumed per ADPN produced and one AN per PN produced, giving:

$$N_{\text{AN}}|_{x=0} = -\frac{2j_1 + j_2}{2F}$$

ADPN and PN are produced at the surface and diffuse away:

$$N_{\text{ADPN}}|_{x=0} = +\frac{j_1}{2F}$$

$$N_{\text{PN}}|_{x=0} = +\frac{j_2}{2F}$$

Hydroxide is produced by all three reactions (2 OH^- per 2 electrons transferred):

$$N_{\text{OH}^-}|_{x=0} = +\frac{j_1 + j_2 + j_3}{F}$$

All other species are non-electroactive and have zero flux:

$$N_i|_{x=0} = 0 \quad \text{for } i \in \{H^+, \text{ phosphates}\}$$

The electrolyte potential $\phi_l(0)$ is **not prescribed** — it is determined by the current conservation equation (§3.2). The overpotentials in the Tafel rate expressions use this solved value.

7.2 Bulk ($x = \delta$): Dirichlet

At the bulk boundary, all species are fixed at their well-mixed bulk values and the electrolyte potential is referenced to zero:

$$c_i(\delta) = c_{i,\text{bulk}} \quad \phi_e(\delta) = 0$$

The bulk equilibrium concentrations of H^+ , OH^- , and the three phosphate species are computed by `solve_phosphate_equilibrium()` (§6.4). The $\phi_l(\delta) = 0$ Dirichlet also serves as the gauge fix required for well-posedness of the current conservation equation (see §3.2).

ADPN and PN bulk concentrations are set to small seed values ($\sim 10^{-3}$ mol m^{-3}) to avoid $\log(0)$ in the initial guess — in an ideal well-mixed bulk they would be zero.

The bulk aqueous AN concentration $c_{AN,\text{bulk}}$ depends on ϵ_{org} because the organic phase sequesters AN. The total amount of AN added to the system partitions between the aqueous phase (concentration $c_{AN,\text{aq}}$) and the organic phase (concentration $c_{AN,\text{org}} = m_{AN} \times c_{AN,\text{aq}}$). Two regimes arise:

Below solubility ($\epsilon_{org} < 0.09$): All AN dissolves in the aqueous phase. The aqueous volume is reduced by the organic fraction, so the dissolved concentration is the total AN divided by the aqueous volume fraction:

$$c_{AN,\text{bulk}} = \frac{C_{AN,\text{total}}}{1 - \epsilon_{org}}$$

where $C_{AN,\text{total}}$ is the moles of AN per unit total volume.

Above solubility ($\epsilon_{org} \geq 0.09$): The aqueous phase is saturated. Additional AN remains in the organic phase as undissolved droplets. The aqueous concentration is pinned at the equilibrium solubility set by the partition coefficient:

$$c_{AN,\text{bulk}} = c_{AN,\text{eq}} = \frac{\rho_{AN}}{M_{AN} m_{AN}} \approx 1,310 \text{ mol } m^{-3}$$

Above the solubility limit, increasing ϵ_{org} does **not** increase $c_{AN,\text{bulk}}$ — it is fixed at $\sim 1,310$ mol m^{-3} . The effect of higher ϵ_{org} enters entirely through $D_{i,\text{mix}}$: the organic droplets provide a

parallel transport pathway for AN, increasing the effective flux of AN toward the electrode without changing the boundary concentration.

8. Three-Parameter Sweep

Parameter	Range	Points
V (cathode potential)	−1.0 to −2.5 V vs SHE	~100 (Newton continuation)
δ (boundary layer thickness)	10, 20, 50, 100, 200 μm	5
ϵ_{org} (organic volume fraction)	0.00, 0.05, 0.09, 0.15, 0.25, 0.30	6

Total: 30 Newton continuation sweeps → 3D performance map FE_ADPN(V, δ , ϵ_{org}).

The diffusion-only limiting current is:

$$j_{\text{lim}} = \frac{n_e F D_{\text{AN,mix}} c_{\text{AN,bulk}}}{\delta}$$

Under the local-equilibrium assumption with the arithmetic-mean D_{mix} , the enhancement comes purely from $D_{\text{AN,mix}} > D_{\text{AN,aq}}$. With the m_i -corrected D_{eff} (§4.2), the enhancement would be much larger. Comparing model predictions under both assumptions against Bloomquist data will determine which is correct.

9. Parameter Tables

9.1 Physical Constants

Symbol	Value	Unit
F	96,485.332	C mol ^{−1}
R	8.314463	J mol ^{−1} K ^{−1}
T	298.15	K (25 °C)

9.2 Kinetic Fitting Parameters

Parameter	Initial	Range	Unit	Basis
E^0_1 (ADPN)	−1.3	−1.0 to −1.5	V vs SHE	Onset (Mathison JACS 2025)

Parameter	Initial	Range	Unit	Basis
E^0_2 (PN)	−1.3	−1.0 to −1.5	V vs SHE	Same reaction centre
E^0_3 (HER)	−0.83	Fixed	V vs SHE	Nernst at pH 14
$j_{0,1}$	10^{-4}	10^{-6} to 10^{-2}	mA cm^{-2}	Irreversible organic reduction
$j_{0,2}$	10^{-4}	10^{-6} to 10^{-2}	mA cm^{-2}	Similar to R1
$j_{0,3}$ (Cd)	10^{-6}	10^{-8} to 10^{-4}	mA cm^{-2}	High HER overpotential
$\alpha_{c,1}$	0.5	0.3–0.7	—	Single-electron RDS
$\alpha_{c,2}$	0.5	0.3–0.7	—	Single-electron RDS
$\alpha_{c,3}$	0.4	0.3–0.5	—	Typical for Cd

Multiply mA cm^{-2} by 10 to convert to A m^{-2} for use in the rate expressions.

9.3 Electrolyte Composition (Modestino Group)

Component	Concentration	Role
Na_3PO_4	0.5 M (500 mol m^{-3})	Buffer + supporting electrolyte
TBA-OH	0.02 M (20 mol m^{-3})	Selectivity agent — lumped into Na^+ (see §2)
EDTA disodium	0.03 M	Metal chelator — not tracked (negligible vs. phosphate)
Acrylonitrile	0.05–0.30 vol fraction	Reactant (sweep parameter)
Cathode	Cd foil	High hydrogen overpotential

9.4 Partition Coefficients (for future m_i upgrade)

Species	$m_i = c_{\text{org}}/c_{\text{aq}}$	Source
AN	11.59	Suwanvaipattana 2017
SN	7.72	Suwanvaipattana 2017
Trimer	30	Suwanvaipattana 2017

Species	$m_i = c_{\text{org}}/c_{\text{aq}}$	Source
Ether	19.02	Suwanvaipattana 2017
All ions	0	Insoluble

10. Numerical Methods

10.1 Mesh

Cell-centred finite-volume on a geometrically graded 1D mesh with $N_{\text{mesh}} = 100$ cells. Cells are finest at $x = 0$ (electrode) and coarsen toward $x = \delta$ (bulk). The mesh is parameterised by a **stretch factor** $s = dx_{\text{max}}/dx_{\text{min}}$ (default $s = 10$). Given s and N , the minimum cell width is derived analytically from the geometric series sum:

$$r = s^{1/(N-1)} \quad (\text{common ratio})$$

$$dx_{\text{min}} = \delta \times (r-1) / (r^N - 1)$$

Cell widths: $dx[k] = dx_{\text{min}} \times r^{(k-1)}$ for $k = 1, \dots, N$. This construction guarantees the mesh fills $[0, \delta]$ exactly for any δ — a fixed absolute dx_{min} would not.

```
function make_mesh(N::Int, delta::Float64; stretch::Float64=10.0)
    r      = stretch^(1.0 / (N - 1))
    dx_min = delta * (r - 1.0) / (r^N - 1.0)
    dx     = [dx_min * r^(k-1) for k in 1:N]
    # Cell-centre positions
    x_c     = cumsum(dx) .- dx ./ 2.0
    return (dx=dx, x_c=x_c)
end
```

At $\delta = 50 \mu\text{m}$ with $N = 100$ and $s = 10$: $dx_{\text{min}} \approx 0.10 \mu\text{m}$, $dx_{\text{max}} \approx 1.0 \mu\text{m}$, dx ratio = 10. Across the full δ sweep (10–200 μm), dx_{min} scales proportionally — resolution is always $\delta/100$ at coarsest and $\delta/1000$ at finest. SG flux for charged species; centred differences for neutrals.

10.2 Jacobian Strategy

The Jacobian has **block-tridiagonal structure** with block size 9 (one per DOF) and half-bandwidth $b = 9$ (lower = upper). Total bandwidth = $2b = 18$.

Banded finite-difference with column coloring. The banded structure means that columns separated by more than $2b+1 = 19$ positions never share a non-zero row — they can be perturbed simultaneously. A sequential-column-grouping coloring therefore requires exactly **19 perturbation vectors** to reconstruct the full Jacobian, regardless of N :

```

# Banded FD Jacobian via column coloring
# bandwidth b = 9 (block size), n_colors = 2b+1 = 19
function banded_fd_jacobian!(J_sparse, residual!, u, F0, b=9)
    n = length(u)
    n_colors = 2*b + 1
    for color in 1:n_colors
        du = zeros(n)
        for j in color:n_colors:n
            du[j] = max(1e-7 * abs(u[j]), 1e-9)
        end
        dF = (residual!(u .+ du) .- F0) ./ du[color:n_colors:n]
        # scatter dF back into the appropriate diagonals of J_sparse
        # (store J_sparse as SparseMatrixCSC or BandedMatrix)
    end
end

```

Jacobian cost: 19 residual evaluations per Newton step vs 900 for dense FD → **47× speedup per step**. Solve cost: sparse/banded LU factorisation ($O(N \times b^2)$) vs dense LU ($O(N^3)$) → further ~50× at $N = 100$. Use `SparseArrays.jl` with `LinearAlgebra.lu` or `KLU.jl` for factorisation.

Perturbation sizes: Use $\varepsilon_{\text{conc}} = 10^{-7} \times \max(|u_i|, 1)$ for log-concentration DOFs; $\varepsilon_{\text{phi}} = 10^{-5}$ V for ϕ_l DOFs. Perturbing $u_i = \ln(c_i)$ corresponds to a multiplicative perturbation of c_i — this is more uniform across the wide concentration range (10^{-10} to 1520 mol m^{-3}) than perturbing c_i directly.

10.3 Newton Solver

Newton with row scaling (divide each residual row by its L_∞ norm at the first iteration) and step clamping:

DOF type	Max step size
Log-concentration u_i	5.0 (natural log units)
Electrolyte potential ϕ_l	0.015 V

The ϕ_l clamp is tightened to 0.015 V (from 0.1 V in v3) because the physical ϕ_l variation across the diffusion layer is only a few mV. A 0.1 V clamp allows Newton to take unphysically large potential steps that couple back into the SG fluxes and destabilise the concentration DOFs.

Convergence tolerance: $\|F\|_\infty < 10^{-4}$.

10.4 Continuation Strategy

Simple Newton continuation (no pseudo-arclength). For each $(\delta, \epsilon_{\text{org}})$ pair, sweep V from -1.0 to -2.5 V vs SHE in uniform steps of $ds = 0.05$ V, warm-starting Newton from the previous converged solution:

- If Newton converges in ≤ 5 iterations: increase ds by factor 1.5 (up to $ds_{\text{max}} = 0.20$ V).
- If Newton fails (diverges or exceeds 25 iterations): halve ds and retry from the last converged point.
- If $ds < ds_{\text{min}} = 0.005$ V: report convergence failure and skip this $(V, \delta, \epsilon_{\text{org}})$ point.

PAC (pseudo-arclength continuation) adds significant implementation complexity (augmented system, arclength parameterisation, tangent predictor) for a system with no expected fold bifurcations. This 1D NP diffusion layer is monotone in V — there is no turning point or bistability. Add PAC later only if the mass-transfer-limited regime shows repeated convergence failures with simple continuation.

10.5 Residual Form

Always use the integrated FV form $F[ix] = J_{\text{left}} - J_{\text{right}} + S \times dx$ (residual in $\text{mol m}^{-2} \text{s}^{-1}$), not the divided form $(J_{\text{right}} - J_{\text{left}})/dx + S$. On the graded mesh (dx ratio up to 10:1 at $s = 10$), the divided form makes residual entries $O(1/dx_{\text{min}})$, degrading Jacobian row scaling.

10.6 Continuation Parameters α_{buf} and α_{kin}

Two scalar continuation parameters scale the source/flux terms during bootstrap (§12). Both are 0 at the start (residual is identically zero given the equilibrium initial guess) and ramped to 1 to recover the full physical model:

Parameter	Scales	Effect at $\alpha = 0$	Effect at $\alpha = 1$
α_{buf}	All buffer source terms $R_{\text{buf},i}$	Buffer chemistry off — equilibrium IC is exact	Full buffer kinetics on
α_{kin}	All Faradaic flux densities j_1, j_2, j_3	No current; flat profiles preserved	Full Tafel kinetics on

Implementation: pass α_{buf} and α_{kin} into the buffer and kinetics evaluators as multiplicative scalars on the rate terms.

```

function buffer_sources!(R, c_H, c_OH, c_H2PO4, c_HP04, c_P04, alpha_buf=1.0)
    r1 = k1f - k1r * c_H * c_OH
    r2 = k2f * c_H2PO4 - k2r * c_H * c_HP04
    r3 = k3f * c_HP04 - k3r * c_H * c_P04
    R[1] = alpha_buf * (+r1 + r2 + r3)
    R[2] = alpha_buf * (+r1)
    R[3] = alpha_buf * (-r2)
    R[4] = alpha_buf * (+r2 - r3)
    R[5] = alpha_buf * (+r3)
    R[6:8] .= 0.0
end

function tafel_currents(c_AN_surface, phi_l_surface, phi_s, alpha_kin=1.0)
    eta1 = (phi_s - phi_l_surface) - E0_1
    eta2 = (phi_s - phi_l_surface) - E0_2
    eta3 = (phi_s - phi_l_surface) - E0_3
    j1 = j0_1 * (c_AN_surface / c_ref)^2 * exp(-alpha_c1 * F * eta1 / (R * T))
    j2 = j0_2 * (c_AN_surface / c_ref) * exp(-alpha_c2 * F * eta2 / (R * T))
    j3 = j0_3 * exp(-alpha_c3 * F * eta3 / (R * T))
    return alpha_kin * j1, alpha_kin * j2, alpha_kin * j3
end

```

10.7 Full Residual Assembly

DOF layout: cell-major with 9 DOFs per cell. For cell ix and species $k \in \{1..8\}$, the log-concentration index is $9*(ix-1) + k$; the ϕ_l index is $9*ix$. Helper inlines:

```

@inline conc_dof(ix, k) = 9*(ix-1) + k    # k = 1..8 (log-concentration DOFs)
@inline phi_dof(ix)     = 9*ix            # one per cell

```

Each cell contributes 9 residual rows: 8 species transport equations + 1 current conservation equation. Faces are indexed 1..N+1, where face 1 is the electrode ($x = 0$), face $ix+1$ is between cells ix and $ix+1$, and face N+1 is the bulk ($x = \delta$). The first cell's left-face flux is the Faradaic boundary flux (§7.1); the last cell's residual is replaced by Dirichlet equations (§7.2).

```

function full_residual!(F, u, mesh, eps_org, V, alpha_buf, alpha_kin, c_eq)
    N = length(mesh.dx)
    z = (+1, -1, -1, -2, -3, 0, 0, 0)    # H+ OH- H2PO4- HPO42- PO43- AN ADPN PN

    # ---- Decode DOFs ----
    c = zeros(8, N)

```

```

phi = zeros(N)
for ix in 1:N
    for k in 1:8
        c[k, ix] = exp(clamp(u[conc_dof(ix, k)], -50.0, 50.0))
    end
    phi[ix] = u[phi_dof(ix)]
end

# ---- Faces 2..N: SG fluxes between adjacent cells ----
J = zeros(8, N+1)
for ix in 1:N-1
    dx_face = 0.5*(mesh.dx[ix] + mesh.dx[ix+1])
    for k in 1:8
        D_k = D_mix(k, eps_org)
        J[k, ix+1] = sg_flux(c[k, ix], c[k, ix+1],
                             phi[ix], phi[ix+1], D_k, z[k], dx_face)
    end
end

# ---- Face 1 (x = 0): Faradaic boundary fluxes ----
# Sign: N_i is the +x flux; consumption at electrode → N_i(0) < 0,
#       production at electrode → N_i(0) > 0.
j1, j2, j3 = tafel_currents(c[6, 1], phi[1], V, alpha_kin)
J[1, 1] = 0.0 # H+ (no Faradaic flux)
J[2, 1] = +(j1 + j2 + j3) / F # OH- produced
J[3, 1] = 0.0; J[4, 1] = 0.0; J[5, 1] = 0.0 # phosphates: no flux
J[6, 1] = -(2*j1 + j2) / (2*F) # AN consumed
J[7, 1] = +(j1) / (2*F) # ADPN produced
J[8, 1] = +(j2) / (2*F) # PN produced
# Face N+1 (x = δ) is unused – Dirichlet override below replaces last-cell residual.

# ---- Assemble per-cell residuals ----
R_buf = zeros(8)
for ix in 1:N
    if ix == N
        # Dirichlet override at the bulk cell
        for k in 1:8
            F[conc_dof(ix, k)] = u[conc_dof(ix, k)] - log(bulk_concentration(k, c_eq, eps_org))
        end
    end
end

```

```

    F[phi_dof(ix)] = phi[ix]                # gauge fix  $\phi_l(\delta) = 0$ 
else
    buffer_sources!(R_buf, c[1,ix], c[2,ix], c[3,ix], c[4,ix], c[5,ix], alpha_buf)
    # Species: integrated FV form  $J_{left} - J_{right} + S \cdot dx$ 
    for k in 1:8
        F[conc_dof(ix, k)] = J[k, ix] - J[k, ix+1] + R_buf[k] * mesh.dx[ix]
    end
    # Current conservation:  $\sum z_i (J_{left} - J_{right})$  over charged species ( $k = 1..5$ )
    sum_left = 0.0
    sum_right = 0.0
    for k in 1:5
        sum_left += z[k] * J[k, ix]
        sum_right += z[k] * J[k, ix+1]
    end
    F[phi_dof(ix)] = sum_left - sum_right
end
end
return F
end

```

The Tafel rate evaluation uses $c[6, 1]$ (cell-centred AN at the first cell) as the surface concentration approximation, and the *solved* $\phi[1]$ as the ϕ_l at the electrode for the overpotential $\eta = (V - \phi_l) - E^0$. This couples the Faradaic BC implicitly to the interior solution.

```

function bulk_concentration(k, c_eq, eps_org)
    k == 1 && return c_eq.H
    k == 2 && return c_eq.OH
    k == 3 && return c_eq.H2PO4
    k == 4 && return c_eq.HPO4
    k == 5 && return c_eq.PO4
    k == 6 && return c_AN_bulk(eps_org)
    k == 7 && return 1e-3                # ADPN seed
    k == 8 && return 1e-3                # PN seed
end

const C_AN_SAT = 1310.0                # mol m-3, aqueous saturation concentration of AN
function c_AN_bulk(eps_org; eps_sat=0.09)
    eps_org < eps_sat ? C_AN_SAT / (1.0 - eps_org) : C_AN_SAT
end

```

10.8 Newton Solver

using SparseArrays, LinearAlgebra

```

function newton_solve!(u, residual!; tol=1e-4, max_iter=25, verbose=false)
    n = length(u)
    F0 = zeros(n)
    J = build_banded_pattern(n, 9)           # block-tridiagonal sparsity, block size 9

    for iter in 1:max_iter
        residual!(F0, u)
        normF = maximum(abs.(F0))
        verbose && @info "Newton" iter normF
        normF < tol && return (converged=true, iter=iter, normF=normF)

        banded_fd_jacobian!(J, residual!, u, F0)  # 19 perturbations (§10.2)
        du = -(J \ F0)                          # sparse LU solve

        # Per-DOF-type step clamping (§10.3)
        N_cells = n ÷ 9
        for ix in 1:N_cells
            for k in 1:8
                idx = conc_dof(ix, k)
                du[idx] = clamp(du[idx], -5.0, 5.0)
            end
            idx_phi = phi_dof(ix)
            du[idx_phi] = clamp(du[idx_phi], -0.015, 0.015)
        end
        u .+= du
    end
    return (converged=false, iter=max_iter, normF=maximum(abs.(F0)))
end

function newton_continuation(u0, V_start, V_end, build_residual;
                             ds_init=0.05, ds_min=0.005, ds_max=0.20)
    u = copy(u0)
    V = V_start
    ds = ds_init
    history = Tuple{Float64, Vector{Float64}}{ }[]

```



```

while V ≥ V_end - 1e-12                                # V_end is more negative than V_start
    residual! = build_residual(V)                        # closure capturing mesh, ε_org, δ, α_*, c_eq
    u_trial = copy(u)
    result = newton_solve!(u_trial, residual!)
    if result.converged
        push!(history, (V, copy(u_trial)))
        u = u_trial
        ds = result.iter ≤ 5 ? min(ds * 1.5, ds_max) : ds
        V -= ds                                         # step toward more negative V
    else
        ds /= 2.0
        ds < ds_min && error("Continuation failed at V = $V")
        # do not advance; retry with smaller ds
    end
end
return history
end

```

Bootstrap (Stage 1, §12) calls `newton_solve!` with progressively larger (α_{buf} , α_{kin}) values. Once both reach 1.0, `newton_continuation` is invoked to sweep V . Each continuation step warm-starts from the previous converged solution.

Banded sparsity pattern: `build_banded_pattern(n, 9)` constructs a `SparseMatrixCSC` with non-zeros at every (i, j) where $|i - j| \leq 9$ (the block-tridiagonal structure of block size 9). `banded_fd_jacobian!` then writes the 19 column-grouped FD perturbations into the existing pattern's non-zeros, avoiding repeated allocation.

11. Solution Caching

Binary files encoding ϵ_{org} , δ , and V in the filename: `s_eo0.150_d50_V-1.300000.bin`. Format: Int64 DOF count + Float64 vector. Cache immediately after Newton converges.

```

function save_solution(u::Vector{Float64}, eps_org, delta_um, V)
    fn = @sprintf("s_eo%.3f_d%.5f_V%.6f.bin", eps_org, round{Int, delta_um*1e6}, V)
    open(joinpath(CACHE_DIR, fn), "w") do f
        write(f, Int64(length(u)))
        write(f, u)
    end
end

```

```

function load_solution(eps_org, delta_um, V)
    fn = @sprintf("s_eo%.3f_d%d_V%.6f.bin", eps_org, round(Int, delta_um*1e6), V)
    path = joinpath(CACHE_DIR, fn)
    isfile(path) || return nothing
    open(path, "r") do f
        n = read(f, Int64)
        read!(f, Vector{Float64}(undef, n))
    end
end

```

12. Implementation Stages (All Mandatory)

Stage 1: NP + migration + buffer + kinetics at $\varepsilon_{\text{org}} = 0$.

Single-phase reference case. No organic droplets, $D_{i,\text{mix}} = D_{i,\text{aq}}$. Validates migration, SG scheme, buffer chemistry, and electrode kinetics.

Sub-steps: 1. Compute $c_{\text{eq}} = \text{solve_phosphate_equilibrium}()$ and build $u0 = \text{make_initial_guess}(\dots)$ (§6.4). At $\alpha_{\text{buf}} = 0$, $\alpha_{\text{kin}} = 0$, residual is identically zero — Newton converges in 0 iterations. 2. **Buffer ramp:** increase α_{buf} from 0 $\rightarrow$ 1 in **10 uniform steps** ($\Delta\alpha = 0.1$) at $V = -1.0$ V vs SHE, with $\alpha_{\text{kin}} = 0$. Each step warm-starts from the previous converged solution. Buffer chemistry is smooth ($Da \gg 1$, no thin reaction zones), so 10 steps is more than sufficient. 3. **Kinetics ramp:** increase α_{kin} geometrically — multiply by 2 each step — starting at $\alpha_{\text{kin}} = 10^{-6}$ and ending at $\alpha_{\text{kin}} = 1.0$ (about **20 steps**: $10^{-6}, 2 \times 10^{-6}, 4 \times 10^{-6}, \dots, 0.5, 1.0$). Still at $V = -1.0$ V. The geometric ramp handles the large dynamic range of kinetic fluxes without requiring hundreds of steps. 4. **Newton continuation sweep over V** from -1.0 to -2.5 V (see §10.4 for step-size control). 5. Cache $\rightarrow$ plots $\rightarrow$ **STOP for review**.

Stage 2: Activate $D_{i,\text{mix}}(\varepsilon_{\text{org}})$.

Same equations, but $D_{i,\text{mix}}$ now depends on ε_{org} . Run at multiple ε_{org} values. This introduces the multiphase transport enhancement (and ionic penalty). Compare polarisation curves and FE_ADPN to Stage 1 to quantify the D_{mix} effect.

Sub-steps: for each ε_{org} , warm-start from Stage 1 solution at $V = -1.0$ V $\rightarrow$ Newton continuation sweep over V $\rightarrow$ cache $\rightarrow$ plots $\rightarrow$ **STOP**.

Stage 3: Full 3D sweep.

Runs Stage 2 across the complete $(\varepsilon_{\text{org}}, \delta)$ grid. 30 Newton continuation sweeps. Generate performance map and all comparison plots $\rightarrow$ **STOP**.

13. Physicality Checks

Check	Expected
$\phi_l(x)$ profile	A few mV variation; larger at high j
$\sum z_i J_i$ at each face	= 0 to machine precision
Electroneutrality	$ \sum z_i c_i < 10^{-8} \text{ mol m}^{-3}$
$c_{\text{AN}}(x)$ at high j	Depletes toward electrode
Buffer residuals at OCV	$< 10^{-6} \text{ mol m}^{-3} \text{ s}^{-1}$
FE_ADPN vs ϵ_{org}	Should increase (enhanced AN transport)
$D_{\text{AN,mix}}$ vs ϵ_{org}	Increases (organic pathway)
$D_{\text{OH,mix}}$ vs ϵ_{org}	Decreases (volume exclusion)
No R_PT residual	Confirm R_PT is absent from residual
Bulk pH at $x = \delta$	~ 13.0 (matches solve_phosphate_equilibrium)

14. Module Structure

```

an_ehd/
├─ params.jl          # Constants (T, F, R, K_w, K_a2, K_a3, D_aq, D_org), Ref-
backed kinetics
├─ mesh.jl            # make_mesh(N, delta; stretch) – stretch-factor graded mesh
├─ diffusivity.jl     # D_mix(i, eps_org), with m_i upgrade hook
├─ chemistry.jl       # solve_phosphate_equilibrium, buffer_sources!, make_initial_guess
├─ kinetics.jl        # j_ADPN, j_PN, j_HER
├─ transport.jl       # sg_flux
├─ assembly.jl        # full_residual! (no R_PT); integrated FV form
├─ solver.jl          # newton_solve! (banded FD Jacobian, sparse LU); newton_continuation
├─ run_stage1.jl      # Single-phase reference → STOP
├─ run_stage2.jl      # D_mix(ε_org) sweep → STOP
├─ run_stage3.jl      # Full 3D sweep → STOP
├─ plot_results.py
└─ output/cache/

```

Note: solve_phosphate_equilibrium and make_initial_guess live in chemistry.jl alongside buffer_sources!. solver.jl uses SparseArrays.jl + KLU.jl (or LinearAlgebra.lu with banded storage) — **no dense Jacobian**. No phase_transfer.jl module — R_PT is absent from this formulation.

15. Common Pitfalls

Pitfall	Symptom	Fix
Using D_{aq} everywhere $D_{org} \neq 0$ for ions	ϵ_{org} has no effect Unphysical ionic transport	Use $D_{mix}(\epsilon_{org})$ Set $D_{org} = 0$ for all ions
Adding ϵ_{aq} to R_{buf}	Double-counting	No prefactor (c_i is aqueous)
Including R_{PT}	Inconsistent with local equilibrium	Remove R_{PT} entirely
Wrong AN order	FE insensitive to j	c^2 for ADPN, c^1 for PN
Cold start (no equilibrium IC)	Newton diverges immediately	Use <code>solve_phosphate_equilibrium</code> + <code>make_initial_guess</code>
SG overflow	exp crash	Clamp α to $[-700, 700]$
Arithmetic D too weak	FE enhancement too small	Upgrade to m_i -corrected D_{eff} (§4.2)
Divided FV form on graded mesh	Jacobian ill-conditioned (entries $O(1/dx_{min})$)	Use integrated form $J_L - J_R + S \cdot dx$ (§10.5)
Missing ϕ_l gauge fix	Jacobian rank-deficient, Newton stalls	Set $\phi_l(\delta) = 0$ (Dirichlet, §7.2)
j_0 in mA/cm ² passed directly	Rates off by factor of 10	Convert: $A/m^2 = mA/cm^2 \times 10$
Dense FD Jacobian	~ 60 ms/step, full sweep > 10 min	Banded FD with 19-column grouping + sparse LU (§10.2)
Fixed absolute dx_{min}/dx_{max} in mesh	Domain length wrong across δ sweep	Use stretch-factor parameterisation via <code>make_mesh</code> (§10.1)
ϕ_l step clamp at 0.1 V	Over-large potential steps destabilise c_i DOFs	Clamp ϕ_l steps to 0.015 V (§10.3)

Pitfall	Symptom	Fix
Implementing PAC before testing Newton cont.	Unnecessary complexity; no fold bifurcations expected	Use simple Newton continuation first (§10.4)

16. Potential Referencing

Model works internally in V vs SHE. Onsets from Mathison et al. (JACS 2025) on Cd: AN reduction -1.28 V; optimal ADPN -1.62 V; HER (no AN) -1.43 V.

Reference electrode	Conversion
Ag/AgCl (sat. KCl)	$E_{\text{SHE}} = E + 0.197 \text{ V}$
SCE	$E_{\text{SHE}} = E + 0.241 \text{ V}$
RHE	$E_{\text{SHE}} = E - 0.059 \times \text{pH} (25^\circ\text{C})$

17. Experimental Data

Bloomquist et al. (CEJ 2026, 528, 172125): $j = 70\text{--}300 \text{ mA cm}^{-2}$, $\varepsilon_{\text{org}} = 0.05\text{--}0.30$. FE_ADPN = 73–76% at $j > 200 \text{ mA cm}^{-2}$. Bubble convection dominates.

Mathison et al. (JACS 2025, 147, 4296): Mechanism — radical coupling for ADPN, proton transfer for PN (strong KIE).

Suwanvaipattana et al. (J. Cleaner Prod. 2017, 142, 1296): D , m , d_p values.

Huang et al. (CEJ 2020, 382, 123006): Langmuir-adsorption kinetics on Pb.

Costentin & Savéant (J. Electroanal. Chem. 564, 2004, 99): $\Delta G^0 = -1.84 \text{ eV}$ for protonated radical coupling.

18. Fitting Strategy

1. $\varepsilon_{\text{org}} = 0$, $\delta = 50 \mu\text{m}$. Fit kinetic parameters (j_0 , α_c) to single-phase FE data.
2. **Activate D_{mix} at $\varepsilon_{\text{org}} = 0.15$.** Check whether FE_ADPN increases. If not: the arithmetic D_{mix} is too weak $\rightarrow$ upgrade to m_i -corrected D_{eff} .
3. **δ sweep at $\varepsilon_{\text{org}} = 0.15$.** Match j_{total} trends to Bloomquist data.

4. ϵ_{org} sweep at fixed δ . Match FE vs organic loading curve.
5. **Full 3D sweep.** Generate performance map.

Metric	Target
FE_ADPN peak ($\epsilon_{\text{org}} = 0$)	50–60%
FE_ADPN peak ($\epsilon_{\text{org}} = 0.15$)	73–80%
FE_ADPN at $j = 200 \text{ mA cm}^{-2}$	> 70%
FE enhancement from $\epsilon_{\text{org}} = 0 \rightarrow 0.15$	+20–30 pp

If the arithmetic D_{mix} cannot produce this enhancement, the m_i correction (§4.2) is needed.

19. Required Plots

Panel	Content	Axes
(a)	$j_{\text{ADPN}}, j_{\text{PN}}, j_{\text{HER}}$ vs V	$-V$ vs SHE / mA cm^{-2}
(b)	$\text{FE}_{\text{ADPN}}, \text{FE}_{\text{PN}}, \text{FE}_{\text{HER}}$ vs V	$-V$ vs SHE / %
(c)	FE_{ADPN} vs j at multiple ϵ_{org}	j / FE — key Bloomquist comparison
(d)	FE_{ADPN} vs ϵ_{org} at fixed j	$\epsilon_{\text{org}} / \text{FE}$
(e)	$c_{\text{AN}}(0)/c_{\text{AN,bulk}}$ vs j	$j / \text{depletion ratio}$
(f)	$\phi_l(0)$ vs j at multiple ϵ_{org}	$j / \text{mV (ohmic penalty)}$
(g)	$D_{\text{AN,mix}}$ and $D_{\text{OH,mix}}$ vs ϵ_{org}	$\epsilon_{\text{org}} / D$
(h)	Production rate vs ϵ_{org}	$\epsilon_{\text{org}} / \text{kg cm}^{-2} \text{ h}^{-1}$

References: Bloomquist et al. CEJ 2026; Corpus et al. Joule 2023; Weng, Bell & Weber PCCP 2018; Huang et al. CEJ 2020; Suwanvaipattana et al. J. Cleaner Prod. 2017; Mathison et al. JACS 2025; Costentin & Savéant J. Electroanal. Chem. 2004; Lasia J. Electroanal. Chem. 1995.